

Skilling Experiment 1: Titanic Survival



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Machine Learning (25SC2107E)

- Cell 1 — Which Library Does What
- Cells 2–3 — Loading and Structure
- Cells 4–5 — Missing Values
- Cells 6–8 — The Survival Plots
- Cell 9 — The Correlation Heatmap
- Deliverable — The EDA Report
- Mixed Review

MCQs — Libraries

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`titanic["survived"].value_counts()`

Answer: (b)

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Answer: (a), (c), (d)

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 (a) the fast array maths underneath pandas (b) drawing the heatmap (c) supplying `np.nan`, the
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- `sns.load_dataset("titanic")` obtains the data by
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- 2. `titanic.shape` prints (891, 15). The number **15** is the count of
 - (a) rows
 - (b) columns
 - (c) missing values
 - (d) input features for the model

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3. Which single command shows **dtypes and non-null counts together**?
- (a) `describe()` (b) `head()` (c) `info()` (d) `shape`

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Baseline

Now imagine the laziest possible "model" — one that ignores age, sex, class, fare, everything, and answers "died" for every single passenger:

- It is right for all 549 who died.
- It is wrong for all 342 who survived.

$$Accuracy = \frac{549}{891} \times 100 = 62\%.$$

MCQs — Counting the gaps

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- ❹ [select ALL that apply] To write **text** onto a blank PDF page you use
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 - (c) `sns.heatmap()`
 - (d) `plt.hist()`

Answer: (a), (b)

MCQs — Why a report at all

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